

# CAMOUFLAGE TECHNIQUE BASED MULTIFUNCTIONAL ARMY ROBOT



RECONNAISSANCE



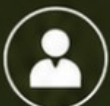
WIRELESS  
CONTROL



CAMOUFLAGE



MULTIFUNCTIONAL  
OPERATIONS



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# ABSTRACT

This project proposes an integrated military robot system built around the ESP32 microcontroller platform. The robot incorporates multiple sensors and capabilities designed for reconnaissance, threat detection, and engagement functions. The system utilizes ultrasonic sensors and metal detectors for environmental awareness and potential threat identification. A 360-degree camera surveillance system provides visual intelligence with integrated face recognition capabilities. The communication architecture employs Zigbee protocols to create a secure link between the robot and a base station laptop.

Defensive and offensive capabilities include a relay-controlled laser targeting system with missile detection algorithms. The complete package offers autonomous operation with manual override options through the base station interface. This robot design balances size, power consumption, and capabilities to provide an effective field-deployable unit suitable for perimeter security, reconnaissance, and limited engagement scenarios.

The modular architecture allows for future capability expansion and sensor upgrades as mission requirements evolve. This project focuses on developing a Military Robot using ESP32, designed for autonomous surveillance, threat detection, and defense operations. The robot is equipped with a metal sensor to detect hazardous objects like landmines or weapons and an ultrasonic sensor for obstacle detection. A laptop camera enables 360-degree surveillance, face recognition, and missile detection, providing real-time intelligence for security personnel.

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## CHAPTER 1

### INTRODUCTION

In modern warfare and border security, the safety of soldiers and the effectiveness of surveillance systems are critical concerns. With rapid advancements in embedded systems, computer vision, and communication technologies, the integration of autonomous or semi-autonomous robots into defines applications has become increasingly practical and impactful. This project proposes a Military Robot designed for surveillance, threat detection, and response, using the ESP32 microcontroller as the core of the system. The robot is equipped with a 360-degree surveillance system using a laptop camera integrated with face recognition and missile detection capabilities. These advanced features allow the robot to identify intruders or known threats and react accordingly in real time. To detect hidden explosives or weapons, a metal sensor is installed, while an ultrasonic sensor ensures effective navigation and obstacle avoidance on the field. For defines and response actions, the robot is fitted with a laser gun controlled via a relay, capable of aiming and simulating fire at detected threats. Alerts are provided using a buzzer, and the system communicates wirelessly with a command centre (laptop) using Zigbee communication for long-range and secure data transfer. All real-time statuses and alerts are shown on an LCD display mounted on the robot. The robot can be deployed in high-risk zones like border areas, restricted military camps, or conflict-prone regions where human presence is risky. This system reduces the exposure of soldiers to direct threats and enhances operational efficiency with constant monitoring, autonomous threat detection, and the ability to engage if necessary.

In an era where technological advancements continue to reshape military operations, autonomous robotic systems have emerged as crucial assets for modern defines strategies. This project presents a comprehensive approach to developing a military-grade robotic system utilizing commercially available components and open-source technologies. The proposed military robot integrates an ESP32 microcontroller as its central processing unit, orchestrating a suite of sensors and actuators to perform surveillance, detection, and defensive functions. By combining ultrasonic sensors, metal detection capabilities, and camera-based surveillance with face recognition algorithms, the robot delivers multi-layered situational awareness in field conditions. Communication between the robot and command centre is facilitated through Zigbee protocols, creating a reliable, secure data link that enables

real-time intelligence sharing and remote operation when required. The integrated LCD interface provides on-unit status information for field personnel interacting directly with the device. The system's defensive capabilities include a precision laser targeting system controlled via relay switching, coupled with missile detection algorithms that can identify and track incoming threats. These features are complemented by auditory alert mechanisms through an integrated buzzer system. This introduction outlines the fundamental architecture of the proposed military robot, setting the foundation for a detailed examination of its components, functionalities, and potential applications in both reconnaissance and defensive scenarios. The following sections will explore the technical implementation, operational parameters, and performance expectations of this integrated robotic platform.

## 1.1 OBJECTIVE

- ❖ To integrate face recognition for identifying authorized and unauthorized personnel.
- ❖ To implement missile detection and automatic destruction using a laser-based defense system.
- ❖ To enable threat detection through metal sensors (for weapons, landmines) and ultrasonic sensors (for objects and obstacles).
- ❖ To establish secure communication between the robot and the control station using Zigbee technology.
- ❖ To provide remote manual control and automatic defense mechanisms.

## 1.2 PROBLEM STATEMENTS

In hostile and high-risk environments such as border areas, war zones, and military installations, human surveillance and threat response put the lives of soldiers in constant danger. Traditional patrolling methods are limited in their capacity to detect hidden threats such as landmines, weapons, or intruders, especially under low-visibility conditions or during continuous monitoring over large areas. There is a critical need for an intelligent, autonomous robotic system that can perform 24/7 surveillance, identify potential threats like unauthorized personnel or incoming missiles, and respond appropriately without risking



human life. Furthermore, current systems often lack real-time communication capabilities, object recognition, or autonomous detection-response mechanisms. This project addresses these challenges by designing a multi-functional military robot equipped with: 360-degree camera surveillance, Face recognition to differentiate between friend and foe, Metal detection for hidden landmines or weapons, Missile detection and response system, Wireless communication using Zigbee, and an automated laser-firing mechanism for defines.

## CHAPTER 2

### LITERATURE REVIEW

#### **1.Sustainability enabling research methods for development of integrated border security systems.**

One of the main priorities for an effective and sustainable Security Union refers to the improvement of information exchange, strengthening information systems and enhancing security at the external borders of the European Union (EU). Recent research projects financed at EU level under Security theme calls are approaching the development of modern technologies and concepts, either for border surveillance and control, or for enhancing IT&C interoperability and common sharing environment enabling for law enforcement agencies. This paper addresses specific new Research Development (R&D) projects types (i.e. pre-operational validation and pre-commercial procurement), aiming to validate them as sustainability enabling research methods for the development of integrated border security systems.

**Date of Conference:** 29 June-1 July 2017

**Date Added to IEEE *Xplore*:** 07 December 2017

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#### **2.Large scale border security systems modeling and simulation with OPNET.**

Most aliens who manage to get across borders undetected are known to end up in cities, and comprise some of the greatest security challenges a country can face. Safety in cities requires critical security measures that ensure that no aliens operate from across the border to avoid local enforcement.

To control the border effectively, countries must deploy an effective detection system that enables real-time surveillance of border integrity. Events such as border



crossings need to be monitored in real time so that any border entries can be noted by border security forces and destinations marked for apprehension.

**Date of Conference:** 9-11 Jan. 2017

**Date Added to IEEE *Xplore*:** 02 March 2017

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### **3. Maritime Border Security using Sensors, Processing, and Platforms to Detect Dark Vessels**

Maritime security is critical to national prosperity, but with a large area to be secured and limited resources available, our current maritime situational awareness is not sufficient to secure our borders. This challenge is compounded by the fact that many bad actors intentionally avoid using transponders to obscure their location and actions for nefarious purposes. New solutions are required to detect these “dark” vessels in the expansive maritime domain and thereby enable increased maritime situational awareness and security. In this paper, we present a concept for increased maritime situational awareness, specifically designed to secure our borders against dark vessels using a novel combination of existing technologies. Our concept is Sensors and Platforms for Unmanned Detection of Dark Ships (SPUDDS), which consists of our intelligent ship detection and classification software integrated onboard our autonomous long-duration sensor buoy, which provides long-range passive detection of non-emitting dark vessels for maritime situational awareness.

**Date of Conference:** 23-24 Oct. 2018

**Date Added to IEEE *Xplore*:** 13 December 2018

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#### **4. Towards Mobile Contactless 4-Fingerprint Authentication for Border Control**

In the last years the importance of biometric authentication in border control procedures increased in a way that biometrics have become the core of most border management systems. Current commercial products for mobile border control have not satisfactorily solved both the demand for increasing security checks and the user requirements driven by security personnel such as border guards yet. Due to their flexibility, portable devices are commonly desired during the control process.

This paper presents on-going work of an advanced mobile device for border control focusing on usability and integrating new technologies to envision next-generation of mobile devices. The device is based on the MobilePass device [13] but significantly improved. A key technology of the new device is a contactless 4-fingerprint authentication instead of only one in existing solutions. Results based on real data shows the advantages of 4-fingerprint versus 1-fingerprint authentication.

**Date of Conference:** 24-25 Oct. 2018

**Date Added to IEEE *Xplore*:** 04 July 2019

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### **5.Smart Border Surveillance System using Wireless Sensor Network and Computer Vision**

Border surveillance is the most important task in the field of national defines and security. To maintain peace and to ensure safety of a country's people, the borders need to be kept under 24/7 monitoring. Especially, under current circumstances, when activities like terrorist infiltrations and illegal movement of both living as well as non-living beings have become common, it becomes of utmost importance to strictly protect the border areas against such activities. To curb such happenings on the border areas, the least that can be done is to provide constant monitoring. In current scenario, this monitoring takes place manually by the border security forces which are responsible for continuously keeping an eye on the borders. It takes a lot of manpower and assets as the borders are stretched across hundreds of miles and have extreme terrain as well as climatic conditions. Hence, the need of the hour is to design an automated border surveillance system which can perform the surveillance task without requiring any human assistance.

**Date of Conference:** 24-26 April 2019

**Date Added to IEEE *Xplore*:** 29 July 2019

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### **6.Intelligent Border Security Intrusion Detection using IoT and Embedded system**

Border areas are generally considered as places where great deal of violence, intrusion and cohesion between several parties happens. This often led to danger for the life of employees, soldiers and common man working or living in border areas. Further geographical conditions like mountains, snow, forest, deserts, harsh weather and water bodies often lead to difficult access and monitoring of border areas. Proposed system uses thermal imaging camera (FLIR) for detection of various objects and infiltrators. FLIR is assigned an IP address and connected through local network to the control centre. Software code captures video and subsequently the intrusion detection. A motor-controlled spotlight with infrared and laser gun is used to illuminate under various conditions at the site. System also integrates sound sensor to detect specific sounds and motion sensors to sense suspicious movements. Based on the decision, a

buzzer and electric current through fence for further protection can be initiated. Sensors are be integrated through IoT for an efficient control of large border area and connectivity between sites.

**Date of Conference:** 15-16 Jan. 2019

**Date Added to IEEE *Xplore*:** 21 February 2019

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## **CHAPTER 3**

### **METHODOLOGY**

#### **SYSTEM REQUIREMENTS AND SPECIFICATION**

##### **Hardware Requirements**

- ❖ Esp32
- ❖ IR sensor
- ❖ Proximity Sensor
- ❖ Laser Gun
- ❖ Relay
- ❖ Zigbee module
- ❖ Power supply

##### **Software Requirements**

- ❖ Arduino IDE
- ❖ Embedded C
- ❖ Python
- ❖ Open cv

## BLOCK DIAGRAM

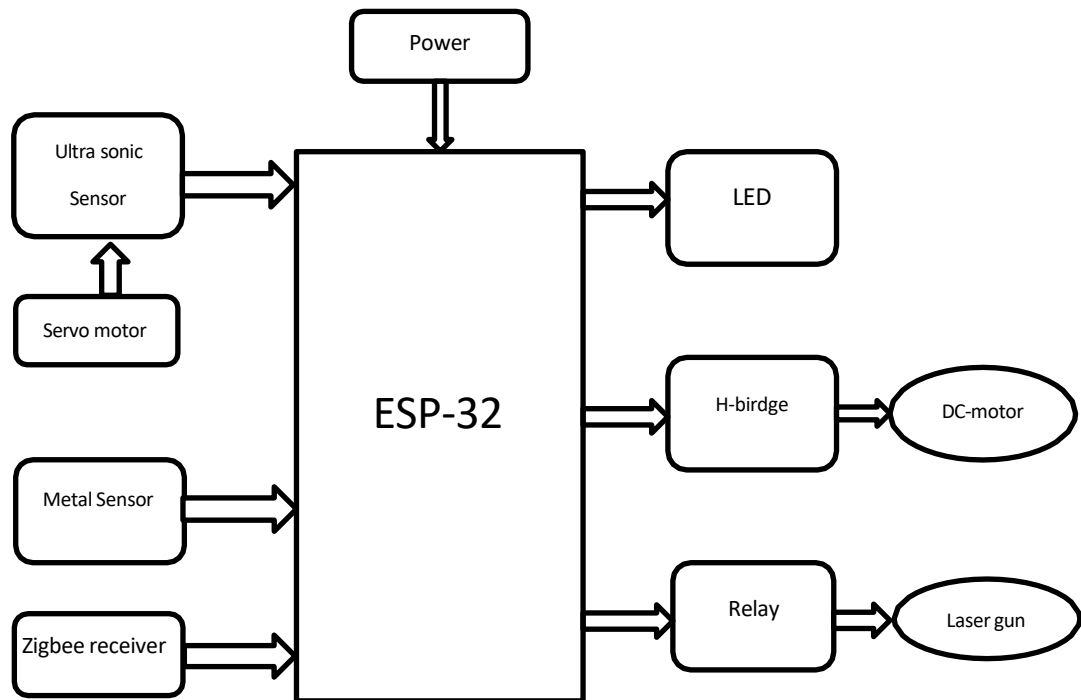


Fig:3.1 Block diagram of Camouflage technique based multifunctional army Robot



Fig:3.2 ZIG-BEE Transmitter

### Hardware Requirements:

#### ESP32:

ESP32 is the SoC (System on Chip) microcontroller which has gained massive popularity recently. Whether the popularity of ESP32 grew because of the growth of IoT or whether IoT grew because of the introduction of ESP32 is debatable. If you know 10 people who have been part of the firmware development for any IoT device, chances are that 7–8 of them would have worked on ESP32 at some point. So, what is the hype all about? Why has ESP32 become so popular so quickly? Let's find out.

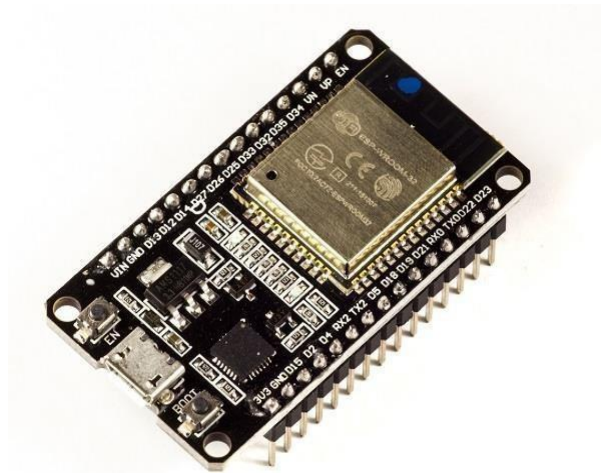


Fig:3.3 ESP-32 micro controller

- ❖ Before we delve into the actual reasons for the popularity of ESP32, let's take a look at some of its important specifications. The specs listed below belong to the ESP32 WROOM 32 variant.–
- ❖ Integrated Crystal– 40 MHz
- ❖ Module Interfaces– UART, SPI, I2C, PWM, ADC, DAC, GPIO, pulse counter, capacitive touch sensor
- ❖ Integrated SPI flash– 4 MB
- ❖ ROM– 448 KB (for booting and core functions)
- ❖ SRAM– 520 KB
- ❖ Integrated Connectivity Protocols– WIFI, Bluetooth, BLE
- ❖ On–chip sensor– Hall sensor
- ❖ Operating temperature range– –40 – 85 degrees Celsius
- ❖ Operating Voltage– 3.3V
- ❖ Operating Current– 80 mA (average)
- ❖ With the above specifications in front of you, it is very easy to decipher the reasons for ESP32's popularity. Consider the requirements an IoT device would have from its microcontroller (act). If you've gone through the previous chapter, you'd have realized that the major operational blocks of any IoT device are sensing, processing, storage, and transmitting.



## Specifications of ESP32

ESP32 has a lot more features than ESP8266 and it is difficult to include all the specifications in this Getting Started with ESP32 guide. So, I made a list of some of the important specifications of ESP32 here. But for complete set of specifications, I strongly suggest you to refer to the Datasheet.

- ❖ Single or Dual-Core 32-bit LX6 Microprocessor with clock frequency up to 240 MHz
- ❖ 520 KB of SRAM, 448 KB of ROM and 16 KB of RTC SRAM.
- ❖ Supports 802.11 b/g/n Wi-Fi connectivity with speeds up to 150 Mbps.
- ❖ Support for both Classic Bluetooth v4.2 and BLE specifications.
- ❖ 34 Programmable GPIOs.
- ❖ Up to 18 channels of 12-bit SAR ADC and 2 channels of 8-bit DAC
- ❖ Serial Connectivity include 4 x SPI, 2 x I<sup>2</sup>C, 2 x I<sup>2</sup>S, 3 x UART.
- ❖ Ethernet MAC for physical LAN Communication (requires external PHY).
- ❖ 1 Host controller for SD/SDIO/MMC and 1 Slave controller for SDIO/SPI.
- ❖ Motor PWM and up to 16-channels of LED PWM.
- ❖ Secure Boot and Flash Encryption.
- ❖ Cryptographic Hardware Acceleration for AES, Hash (SHA-2), RSA, ECC and RNG.

## Different Ways to Program

A good hardware like ESP32 will be more user friendly if it can be programmed (writing code) in more than one way. And not surprisingly, the ESP32 supports multiple programming environments.

Some of the commonly used programming environments are:

- Arduino IDE
- Platform IO IDE (VS Code)
- LUA
- Micro Python
- Espresso IDF (IoT Development Framework)
- JavaScript

As Arduino IDE is already a familiar environment, we will use the same to program ESP32 in our upcoming projects. But you can definitely try out others as well.

### Layout

We will see what a typical ESP32 Development Board consists of by taking a look at the layout of one of the popular low-cost ESP Boards available in the market called the ESP32 Devkit Board.

The following image shows the layout of an ESP32 Development Board which I have.

**IMPORTANT NOTE:** There are many ESP32 Boards based on ESP-WROOM-32 Module available in the market. The layout, pinout and features vary from board to board.

The board which I have has 30 Pins (15 pins on each side). There are some boards with 36 Pins and some with slightly less Pins. So, double check the pins before making connections or even powering up the board.

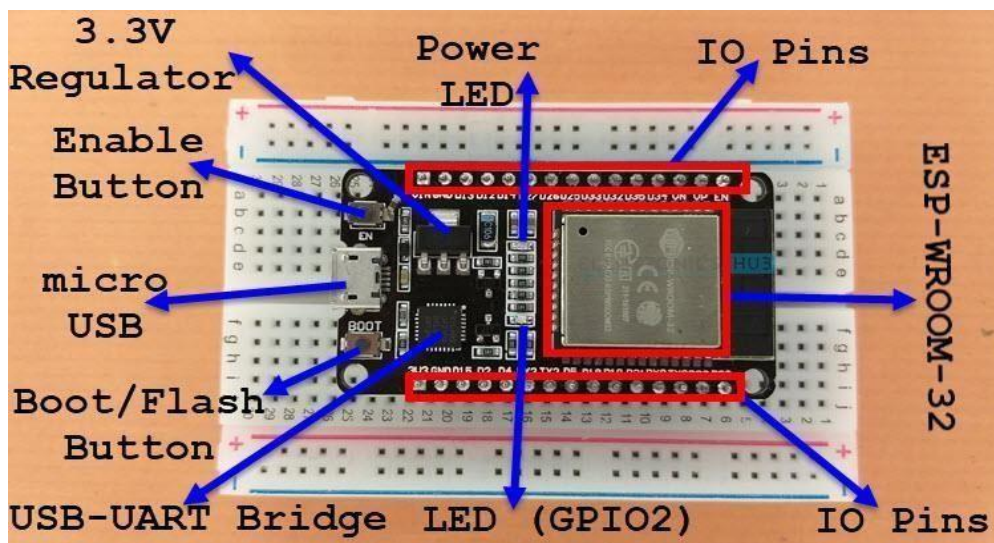


Fig:3.4 Layout of ESP-32 microcontroller

As you can see from the image, the ESP32 Board consists of the following:

These are the contents of the layout of ESP-32 micro controller and how it designed.

#### ❖ ESP-WROOM-32 Module

- ❖ Two rows of IO Pins (with 15 pins on each side)
- ❖ CP2012 USB – UART Bridge IC
- ❖ micro–USB Connector (for power and programming)
- ❖ AMS1117 3.3V Regulator IC
- ❖ Enable Button (for Reset)
- ❖ Boot Button (for flashing)
- ❖ Power LED (Red)
- ❖ User LED (Blue – connected to GPIO2)
- ❖ Some passive components

This pinout is for the 30 – pin version of the ESP Board.

In the pinout tutorial, I will explain the pin out of both the 30 – pin as well as the 36 – pin version of the ESP Boards.

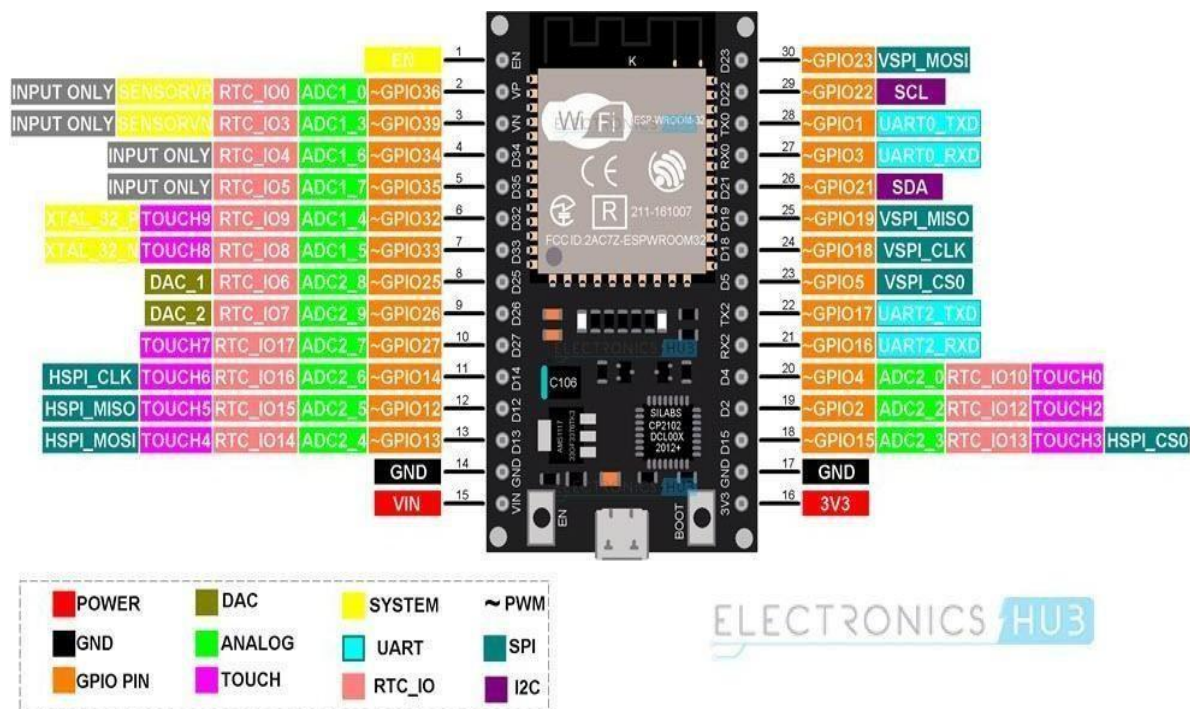


Fig:3.5 PIN configuration of ESP-32 micro controller

## Relay

A relay is an electrically operated switch. Current flowing through the coil of the relay



Fig:3.6 Relay

creates a magnetic field which attracts a lever and changes the switch contacts. The coil current can be on or off so relays have two switch positions and they are double throw (changeover) switches. The relay's switch connections are usually labeled COM(POLE), NC and NO. In order to trigger the laser, we use driver relay.

Many relays use an electromagnet to mechanically operate a switch, but other operating principles are also used, such as solid-state relays. Relays are used where it is necessary to control a circuit by a separate low-power signal, or where several circuits must be controlled by one signal. Relays were used extensively in telephone exchanges and early computers to perform logical operations.

## **DC Motor**

DC motors convert electrical into mechanical energy and they consist of permanent magnets and loops of wire inside, when current is applied, the wire loops generate a magnetic field, which reacts against the outside field of the static magnets. The interaction of the fields produces the movement of the shaft/armature. Thus, electromagnetic energy becomes motion.

Here we use two DC motors for the movement of rover. A DC motor is any of a class of rotary electrical machines that converts direct current electrical energy into mechanical energy. The most common types rely on the forces produced by magnetic fields. Nearly all types of DC motors have some internal mechanism, either electro mechanical or electronic, to periodically change the direction of current flow in part of the motor. DC motors were the first form of motor widely used, as they could be powered from existing direct-current



Fig:3.7 DC-MOTOR

lighting power distribution systems. A DC motor's speed can be controlled over a wide range, using either a variable supply voltage or by changing the strength of current in its field windings.

#### 3.1.4 H-Bridge: L293D:

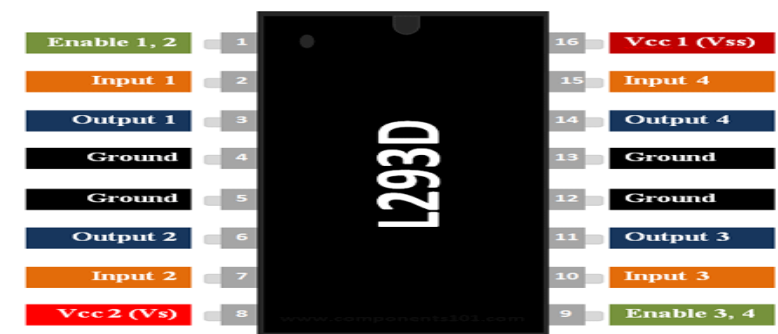


Fig:3.8 H-Bridge for DC motor

#### L293D Pin Configuration

##### Features

- ❖ Can be used to run Two DC motors with the same IC.
- ❖ Speed and Direction control is possible
- ❖ Motor voltage Vcc2 (Vs): 4.5V to 36V
- ❖ Maximum Peak motor current: 1.2A
- ❖ Maximum Continuous Motor Current: 600mA
- ❖ Supply Voltage to Vcc1(vs.): 4.5V to 7V

- ❖ Transition time: 300ns (at 5V and 24V)
- ❖ Automatic Thermal shutdown is available
- ❖ Available in 16-pin DIP, TSSOP, SOIC packages.

### 3.1.5 Buzzer:



Fig:3.9 Buzzer

#### Buzzer Pin Configuration

| Pin Number | Pin Name | Description                                                                         |
|------------|----------|-------------------------------------------------------------------------------------|
| 1          | Positive | Identified by (+) symbol or longer terminal lead. Can be powered by 6V DC           |
| 2          | Negative | Identified by short terminal lead. Typically connected to the ground of the circuit |

#### Buzzer Features and Specifications

- ❖ Rated Voltage: 6V DC
- ❖ Operating Voltage: 4-8V DC
- ❖ Rated current: <30mA
- ❖ Sound Type: Continuous Beep
- ❖ Resonant Frequency: ~2300 Hz
- ❖ Small and neat sealed package
- ❖ Breadboard and Perf board friendly

## How to use a Buzzer

A **buzzer** is a small yet efficient component to add sound features to our project/system. It is very small and compact 2-pin structure hence can be easily used on breadboard, Perf Board and even on PCBs which makes this a widely used component in most electronic applications.

There are two types of buzzers that are commonly available. The one shown here is a simple buzzer which when powered will make a Continuous Beweeps.... sound, the other type is called a readymade buzzer which will look bulkier than this and will produce a Beep. Beep. Beep.

This buzzer can be used by simply powering it using a DC power supply ranging from 4V to 9V. A simple 9V battery can also be used, but it is recommended to use a regulated +5V or +6V DC supply. The buzzer is normally associated with a switching circuit to turn ON or turn OFF the buzzer at required time and required interval.

### Applications of Buzzer

- Alarming Circuits, where the user has to be alarmed about something
- Communication equipment's.
- Automobile electronics
- Portable equipment's, due to its compact size

## Regulated power supply:

### Transformer:

A transformer is a device that transfers electrical energy from one circuit to another through inductively coupled conductors without changing its frequency. A varying current in the first or primary winding creates a varying magnetic flux in the transformer's core, and thus a varying magnetic field through the secondary winding.

### Rectifier:

A rectifier is an electrical device that converts alternating current (AC) to direct current (DC), a process known as rectification. A device that it can perform the opposite



function (converting DC to AC) is known as an inverter.

### **Filter:**

The process of converting a pulsating direct current to a pure direct current using filters is called as filtration. Electronic filters are electronic circuits, which perform signal-processing functions, specifically to remove unwanted frequency components from the signal, to enhance wanted ones.

### **Regulator:**

A voltage regulator (also called a \_regulator ‘) with only three terminals appears to be a simple device, but it is in fact a very complex integrated circuit. It converts a varying input voltage into a constant \_regulated ‘output voltage. Voltage Regulators are available in a variety of outputs like 5V, 6V, 9V, 12V and 15V. The LM78XX series of voltage regulators are designed for positive input. For applications requiring negative input, the LM79XX series is used. Using a pair of \_voltage-divider ‘resistors can increase the output voltage of a regulator circuit.

### **Laser:**

A laser is a device that emits light through a process of optical amplification based on the stimulated emission of electromagnetic radiation. The term "laser" originated as an acronym for “Light Amplification by Stimulated Emission of Radiation”. The term "laser" originated as an acronym for "Light Amplification by Stimulated Emission of Radiation". The first laser was built in 1960 by Theodore Althernatively, temporal coherence can be used to produce pulses of light with a broad spectrum but durations as short as a femtosecond ("ultrashort pulses").



Fig:3.9 Laser



Spatial coherence also allows a laser beam to stay narrow over great distances (collimation), enabling applications such as laser pointers and lidar. Lasers can also have high temporal coherence, which allows them to emit light with a very narrow spectrum, i.e., they can emit a single color of light.

### **ZIGBEE TB 387 2.4 GHZ:**

#### **FEATURES**

- ❖ Working voltage: 3.3V-5.5V;
- ❖ RS232 Interface (3.3V/5V TTL level)
- ❖ Frequency range: 2402~2482MHz
- ❖ Transmit power: 20dBm( 100mw);
- ❖ Receiver sensitivity: -87dBm;
- ❖ Operating temperature: -40~+85 ° c;
- ❖ Transparent transmission mode baud rate:
- ❖ 2400, 4800, 9600(Default),14400, 19200, 38400, 57600, 115200, 12800, 2560
- ❖ AT mode configured baud rate fix is: 9600;
- ❖ Open ground Transmission distance: 500M



Fig:3.10 ZIG-BEE for Communication

#### **Pin Description:**

- 1, VDD: 3.3V or 5V power supply positive
- 2, GND: Power negative
- 3, TXD: serial output module, connected microcontroller or USB to serial RXD
- 4, RXD: module serial input connected microcontroller or USB to serial TXD
- 5, CMD: enter upgrade mode or AT mode pin, active low

## Software Requirements:

### Arduino IDE

A program for Arduino may be written in any programming language for a compiler that produces binary machine code for the target processor. Atmel provides a development environment for their microcontrollers, AVR Studio and the newer Atmel Studio.

The Arduino project provides the Arduino integrated development environment (IDE), which is a cross-platform application written in the programming language Java. It originated from the IDE for the languages Processing and A program written with the IDE for Arduino is called a sketch The Arduino IDE supplies a software library from the Wiring project, which provides many common input and output procedures.

A minimal Arduino C/C++ sketch, as seen by the Arduino IDE programmer, consist of only two functions:



Fig:3.11 Arduino IDE (version 2.3.2)

- **setup ():** This function is called once when a sketch starts after power-up or reset. It is used to initialize variables, input and output pin modes, and other libraries needed in the sketch.
- **loop ():** After setup () has been called, function loop () is executed repeatedly in the main program. It controls the board until the board is powered off or is reset.

**CODE**

```
#include <stdio.h>

#include <stdlib.h>

#include <string.h>

#define heart 13
#include
<LiquidCrystal_I2C.h>
LiquidCrystal_I2C lcd(0x27,16, 2);
#include <ESP32Servo.h>

Servo servo1;int IN1=14;

int IN2=27; int IN4=26;

int IN3=25; int IN7=33;

int IN8=32;

#define trigPin 15

#define echoPin 2

long duration;

int distance;

//int Relay1=4;

//int Relay2=5;

//int Relay3=18; int Laser=19; int Metal=23; char ch;

int pos = 0;

void setup(void)
```

```
{  
  
    servo1.attach(5);  
  
    pinMode(Metal, INPUT);  
  
    servo1.write(0);  
  
    pinMode(IN1, OUTPUT);  
  
    pinMode(IN2, OUTPUT);  
  
    pinMode(IN3, OUTPUT);  
  
    pinMode(IN4, OUTPUT);  
  
    pinMode(IN7, OUTPUT);  
  
    pinMode(IN8, OUTPUT);  
  
    pinMode(trigPin,OUTPUT);  
  
    pinMode(echoPin,INPUT);  
  
    //pinMode(Relay1,OUTPUT);  
    // pinMode(Relay2,OUTPUT);  
    //pinMode(Relay3,OUTPUT);  
  
    pinMode(Laser, OUTPUT);  
  
    digitalWrite(IN1,LOW);  
  
    digitalWrite(IN2,LOW);  
  
    digitalWrite(IN3,LOW);  
  
    digitalWrite(IN4,LOW);  
  
    // digitalWrite(Relay1,LOW);  
  
    // digitalWrite(Relay2,LOW);
```

```
// digitalWrite(Relay3,LOW);
```

```
digitalWrite(Laser,LOW);
```

```
Serial.begin(9600);
```

```
Serial2.begin(9600);
```

```
lcd.init();
```

```
// Initialize the LCD
```

```
lcd.backlight();
```

```
lcd.clear();
```

```
lcd.print("Military Robot..");
```

```
Serial.println("MilitaryRobot.");
```

```
delay(1000);
```

```
}
```

```
void loop(void)
```

```
{
```

```
Check();
```

```
}
```

```
void Check()
```

```
{
```

```
Forward();
```

```
Sensor_Check();
```

```
Waiting();
```

```
}
```

```
void Disposal()

{

for(int x=0;x<5;x++)

{

    lcd.clear();

    lcd.print("Disposing..");

    Serial.println("Disposing..");

    digitalWrite(IN7,HIGH);

    digitalWrite(IN8,LOW);

    delay(1000);

    digitalWrite(IN7,LOW);

    digitalWrite(IN8,LOW);

    delay(1000);

    digitalWrite(IN7,LOW);

    digitalWrite(IN8,HIGH);

    delay(1000);

    digitalWrite(IN7,LOW);

    digitalWrite(IN8,LOW);

    delay(1000);

}

Waiting();

}
```

```
void Ultrasonic_Check()

{

digitalWrite(trigPin, LOW);

delayMicroseconds(5);

// Trigger the sensor by setting
the  trigPin  high  for  10
microseconds:

digitalWrite(trigPin, HIGH);

delayMicroseconds(10);

digitalWrite(trigPin, LOW);
// Read the echoPin, pulseIn()
returns the duration (length of
the pulse) in microseconds:

duration = pulseIn(echoPin,

HIGH);

// Calculate the distance:

distance = duration * 0.034 /2;

// Print the distance on the
Serial Monitor (Ctrl+Shift+M):

// Serial.print("Distance = ");

// Serial.print(distance);

// Serial.println(" cm");

lcd.clear();
```

```
    lcd.print("Distance= ");

    lcd.print(distance);

    lcd.print("cm");

    delay(1000);

    if(distance<50)

    {

        //      Serial.println("$Person
Detected..#");

        lcd.clear();

        lcd.print("Object
Detected..");

        Stop();

        delay(2000);

        //Waiting();

    }

    Waiting();

}

void Waiting()

{

    lcd.clear();

    lcd.print("Waiting..");

    // while(1)

    // {
```



```
if(Serial2.available(>0)

{
    Stringbuff=Serial2.readStrin();

    Serial.println(buff);

    delay(1000);

    if(buff[0]=='$')
    {

delay(500);

if(buff[1]=='A')

{

    lcd.clear();
    lcd.print("Known Person..");

    Serial.println("Known Person");

    delay(500);

    Forward();

    Check();

    // break;

    }
    if(buff[1]=='U')

{

    lcd.clear();

    lcd.print("Unknown Person..");

    Serial.println("Unknown Person");

    delay(500);

    Stop();
    delay(100);

digitalWrite(Laser,HIGH);
```

```
    delay(5000);

    digitalWrite(Laser,LOW);

    Check();

    // break;

}

if(buff[1]=='M')

{
    Serial.println("Missile On..");
    lcd.clear();

    lcd.print("Missile On..");

    servo1.write(90);

    delay(2000);

    servo1.write(0);

    Check();

    //break;
}

// if(buff[1]=='G')

// {
    delay(100);

    digitalWrite(Laser,HIGH);

    delay(5000);

    digitalWrite(Laser,LOW);

    Check();

    // break;
```

```
}

if(buff[1]=='M')

{
    Serial.println("Missile On.");

    lcd.clear();
    lcd.print("Missile On.");
    servo1.write(90);

    delay(2000);

    servo1.write(0);

    Check();

//  break;

}

//if(buff[1]=='G')

// {
//     Serial.println("Green On.");
//     lcd.clear();

//     lcd.print("Green On.");
//     digitalWrite(Relay3,LOW);

//     digitalWrite(Relay1,LOW);

//     digitalWrite(Relay2,HIGH);

//     delay(2000);

//     Check();

//     break;

// }
```

```
// if(buff[1]=='B')

// {

//   Serial.println("Blue On.");

//   lcd.clear();

//   lcd.print("Blue  On..");

digitalWrite(Relay2,LOW);

digitalWrite(Relay1,LOW);

// digitalWrite(Relay3,HIGH);

// delay(2000);

//   Check();

//   break;
// }

//else

// {

//   digitalWrite(Relay2,LOW);

//   digitalWrite(Relay1,LOW);

//   digitalWrite(Relay3,LOW);

// delay(500); } } }

void Forward()

{

  lcd.clear();

  lcd.print("Forward..");
```

```
Serial.println("Forward..");

digitalWrite(IN1,HIGH);

digitalWrite(IN2,LOW);

digitalWrite(IN3,HIGH);

digitalWrite(IN4,LOW);

// Waiting();
}
void Reverse()

{

    lcd.clear();

    lcd.print("Reverse..");

    Serial.println("Reverse..");

    digitalWrite(IN1,LOW);

    digitalWrite(IN2,HIGH);

    digitalWrite(IN3,LOW);

    digitalWrite(IN4,HIGH);

    }

    void Left()
    {

        lcd.clear();

        lcd.print("Left..");

        Serial.println("Left..");

        digitalWrite(IN1,LOW);
```

```
digitalWrite(IN3,HIGH);
```

```
digitalWrite(IN4,LOW);
```

```
    }  
    void Right()
```

```
{  
  lcd.clear();
```

```
  lcd.print("Right..");
```

```
  Serial.println("Right..");
```

```
  digitalWrite(IN1,HIGH);
```

```
  digitalWrite(IN2,LOW);
```

```
  digitalWrite(IN3,LOW);
```

```
  digitalWrite(IN4,HIGH);
```

```
    }  
    void Stop()
```

```
{
```

```
  lcd.clear();
```

```
  lcd.print("Stop..");
```

```
  Serial.println("Stop..");
```

```
  digitalWrite(IN1,LOW);
```

```
  digitalWrite(IN2,LOW);
```

```
  digitalWrite(IN3,LOW);
```

```
  digitalWrite(IN4,LOW);
```

## CHAPTER-5

### RESULTS

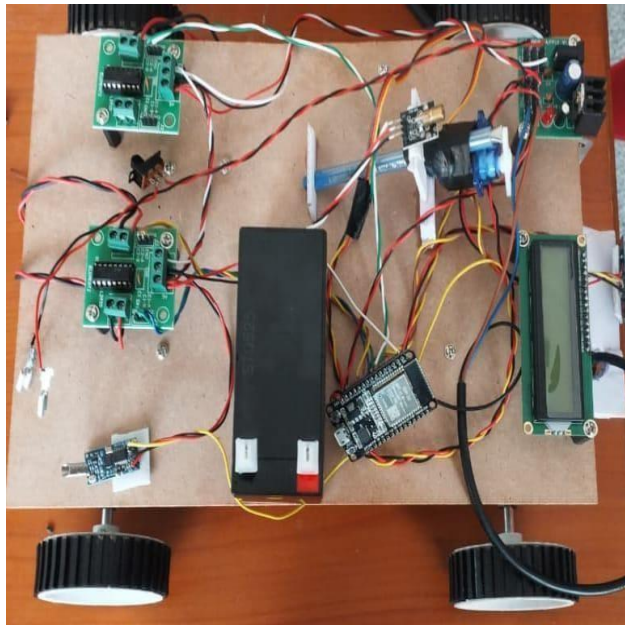


Fig 3.12 Side view

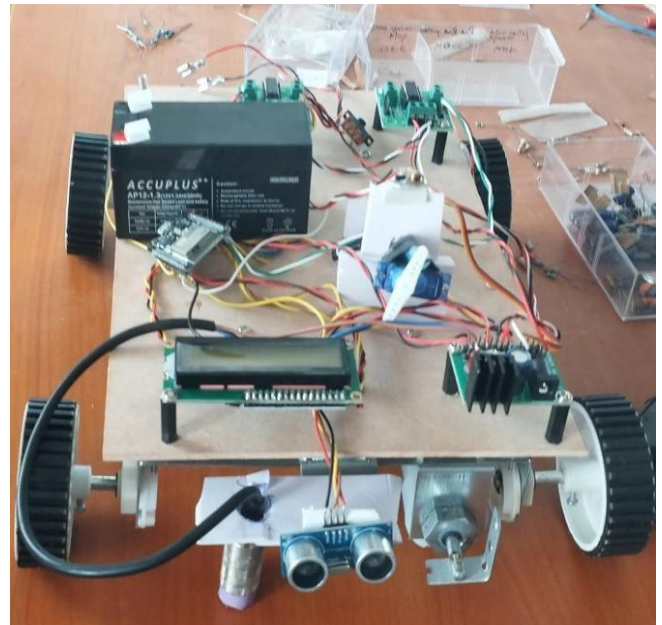


Fig 3.12 Front view

As shown in the figure our robot works based on four applications. In which first one is it measures distance of the object by using the ultrasonic sensor and display the distance object on the LCD display. The second application is face detection or recognitions of the person. The camera captures the image of an object and compares with the built-in image of the known person if unknown person is detected the missile is on for attacking the person (is remote control or automatic). The third application is missile detection missile monitoring takes place through camera if missile is detected the attacking system is on and destroys the missile. The last application is metal digging if any bomb is detected inside the earth, it starts digging.

## **CHAPTER -6**

### **ADVANTAGES AND DISADVANTAGES**

#### **ADVANTAGES**

- This System is an effective and a safe system to ensure that there are no humans left behind in a rescue operation.
- The System is safe even for the user because of the use of robotics and no manual work
- The system uses wifi and this makes the system both accurate and reliable
- Safety and efficiency.
- Large sensing distance.
- Increasing responsiveness and situational awareness.
- Ethical and Technical safety layer.
- Operators direct control when needed.

#### **DISADVANTAGES**

- Human intervention is reduced, but not eliminated.
- System requires considerable initial setup.
- The angle of camera movement is limited
- Speed is slow of robot
- Video quality is ok.

#### **APPLICATIONS**

- Military defense systems for missile and object detection and destruction.
- Used in army, only for border areas.
- Real time monitoring helps for different fields.



## CONCLUSION

The combination of IoT and sonar-based recognition. This ESP32-based Military Robot functions as an intelligent surveillance and defense system, providing real-time threat detection, remote control capabilities, and autonomous defensive actions. The robot also captures live footage of the surrounding area so that we can monetarize and analyse the war field's territory. So, in all the proposed system provides a helping hand to our security forces in detection of intruders. The robot can also be used in high altitude areas where human cannot survive.

The Military Robot using ESP32 is a versatile and intelligent defense system designed to enhance battlefield security through autonomous surveillance, object detection, and threat elimination. By integrating 360-degree surveillance with a laptop camera, face recognition, missile detection, and laser-based defense mechanisms, the system provides a real-time security solution for military applications. The metal and ultrasonic sensors improve its ability to detect obstacles and hazardous objects such as landmines or weapons, ensuring safe navigation in combat zones. Overall, this project demonstrates a cost-effective, scalable, and efficient approach to military surveillance and defense. Future enhancements, such as AI-driven decision-making, GPS-based navigation, cloud integration, and advanced weapon systems, could further expand its capabilities, making it an indispensable asset in modern warfare and security operations.

## FUTURE SCOPE:

Our project is mainly intended to operate a robot using PC. The system also to detects the metallic presence and the PIR sensor detects any human presence in its way and if any human presence is being detected it stops and buzzers an alarm system. The micro controller is programmed in such a way that the depending on the pressed key the robot will move intelligently and with the help of metal detection sensor it detects the presence of metallic objects in mines and alerts through buzzer alarm system. The drawback of this project is that the status of robot is not known. This can eliminate by having a GSM module, which gives the status of robot working. We can also add Ultrasonic module, which is used for obstacle detection with GSM module which gives respective information. By connecting wireless camera to the robot, then we can see the outer world from our personal computer only by using GPRS and GPS. We can use this robot at so many fields and we can use to

handle so many situations. By connecting bomb detector to the robot, we can send it to anywhere i.e. (battle field, forests, coal mines, to anyplace) by using our personal computer and we can able to detect the bomb at field, here sensor detects the bomb and gives information to micro controller and it gives the information to transceiver and it sends the information to the personal computer. By connecting temperature sensor to the robot, we can get the temperature of dangerous zones in personal computer itself instead of sending human to there and facing problems at field we can send robot to there and sensor will detect the temperature and it gives information to the micro controller and micro controller gives the information to the transceiver from that we can get the data at pc side. By connecting smoke sensor to the robot, we can get the information related concentration of smoke or gases in respective field's i.e. (coal mines, dangerous zones, etc.). sensor sense the information and it give to the micro controller and it gives to the transceiver and from that we get the information in personal computer. By connecting corresponding instruments to the robot, we can use it in agriculture for farming purpose. This robot can move either forward and backward and left and right depend upon our instructions so we can do some part of agriculture from pc only by using robot. By connecting firing instrument and wireless camera to the robot we can fire the target from pc. Here by using camera we can see the opposite target and we can fire the target from personal computer by pressing selected button and we can easily handle the situations like Mumbai terrorist's attack without loss of human life's and we can decrease our soldier's effort too.